ANALYSIS REPORT

Track 1 SVR Reference-Grid Repair Campaign Results Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the dedicated `SVR` reference-grid repair campaign prepared in:

- `doc/reports/campaign_plans/2026-04-14-22-53-48_track1_svr_reference_grid_search_repair_campaign_plan_report.md`

  The campaign executed `4` exact-paper validation runs through the stabilized remote launcher:
- completed runs: `4`
- failed runs: `0`
- execution window: `2026-04-17 00:54:27+02:00` to `2026-04-17 06:29:49+02:00`
- campaign artifact root:

  `output/training_campaigns/track1_svr_reference_grid_search_repair_campaign_2026_04_14_22_53_48/` The campaign goal was twofold:
- verify that the remote exact-paper `SVR` `GridSearchCV` launcher now runs

  end-to-end on the LAN workstation;
- rerun the remaining `SVR` paper-gap targets `40`, `240`, and `162` with the

  paper-faithful grid instead of the earlier seed/split sweeps.

Objective And Outcome

The campaign had four concrete questions

1. can the repaired remote exact-paper wrapper complete a real `4/4` campaign

   without hangs or path-resolution failures?
2. does the paper-faithful `SVR` grid close amplitude harmonic `40`?
3. does it close amplitude harmonic `240`?
4. does it close phase harmonic `162`?

   Outcome:

- the remote launcher completed the full campaign end-to-end and synchronized

  all validation artifacts back to the repository;
- all `4` scoped validation runs completed successfully;
- none of the scoped paper targets reached or beat the paper numeric target;
- the isolated runs reproduced the same `SVR` metrics already seen in the

  pair/smoke diagnostics;
- all scoped `SVR` ONNX exports failed under the current stack, but the runs

  still completed because `export_failure_mode` remained `continue`. This means the operational objective succeeded, but the scientific repair objective did not close any new paper cells.

Ranking Policy

This campaign is not a full family-matrix comparison and it is not a true closure campaign, because every run stayed above its scoped paper target.

The serialized campaign policy is therefore

- primary metric: `mean_normalized_gap_ratio_asc`
- first tie breaker: `max_normalized_gap_ratio_asc`
- second tie breaker: `open_paper_cell_count_asc`
- third tie breaker: `failed_target_count_asc`
- fourth tie breaker: `run_name`

Where:

- `normalized_gap_ratio = (repository_metric - paper_metric) / paper_metric`
- lower is better
- the phase run uses both paper cells available on harmonic `162`:

`MAE` and `RMSE`

- the amplitude runs use the paper `RMSE` cells available for harmonics `40`

and `240` Interpretation:

- this policy exists only to serialize the required campaign-best artifacts;
- the scientific reading remains that all scoped `SVR` targets are still open.

Campaign Ranking

Ranked Completed Runs

Rank	Run	Scope	Paper Cells	Mean Gap Ratio	Max Gap Ratio	Failed Exports
1	`track1_svr_reference_grid_phase_162_only`	`phases_only`	2	1.307	1.711	1
2	`track1_svr_reference_grid_amplitude_40_only`	`amplitudes_only`	1	1.676	1.676	1
3	`track1_svr_reference_grid_amplitude_pair`	`amplitudes_only`	2	7.745	13.814	2
4	`track1_svr_reference_grid_amplitude_240_only`	`amplitudes_only`	1	13.814	13.814	1

Export Surface

Completed Runs	Exported ONNX Files	Failed Exports	Surrogate Exports
4	0	5	0

The launcher path is therefore now stable, but the exact-paper `SVR` ONNX export path is still not operational under the current environment.

Campaign Best Run

The explicit bookkeeping winner is

- `track1_svr_reference_grid_phase_162_only`

It was selected because it achieved:

- the lowest scoped mean normalized paper-gap ratio: `1.307`
- the lowest scoped max normalized paper-gap ratio among the four runs
- a cleaner paper-gap approach than the two amplitude `240` variants

Important interpretation:

- this run is the campaign bookkeeping winner;
- it is **not** a closure run;
- both scoped paper cells for phase `162` remain above the paper target.

Scientific Outcome

Scoped Target Metrics

Run	Best Params	Scoped Result
`track1_svr_reference_grid_amplitude_pair`	`C=0.1` , `epsilon=1e-05` , `kernel=rbf`	`A40` RMSE `9.63e-05` ; `A240` RMSE `6.22e-04`
`track1_svr_reference_grid_amplitude_40_only`	`C=0.001` , `epsilon=0.0001` , `kernel=rbf`	`A40` RMSE `9.63e-05`
`track1_svr_reference_grid_amplitude_240_only`	`C=0.1` , `epsilon=1e-05` , `kernel=rbf`	`A240` RMSE `6.22e-04`
`track1_svr_reference_grid_phase_162_only`	`C=1.0` , `epsilon=1e-06` , `kernel=rbf`	`phi162` MAE `0.542246` ; RMSE `1.217833`

Gap Versus Paper

Harmonic / Metric	Paper Target	Repository Result	Gap	Status
`A40` `RMSE`	`3.60e-05`	`9.63e-05`	`6.03e-05`	above paper target
`A240` `RMSE`	`4.20e-05`	`6.22e-04`	`5.80e-04`	above paper target
`phi162` `MAE`	`0.200`	`0.542246`	`0.342246`	above paper target
`phi162` `RMSE`	`0.640`	`1.217833`	`0.577833`	above paper target

What The Campaign Actually Established

- the isolated `40` run does not improve over the already known `SVR` result;
- the isolated `240` run does not improve over the pair run;
- the isolated `162` run remains substantially above the paper target;
- the paper-faithful `SVR` grid therefore does not reveal a hidden target-specific recovery on these residual harmonics.

Operational Outcome

The campaign still delivered a meaningful infrastructure result.

It proves that the remote exact-paper workflow now works end-to-end for a real multi-run campaign:

- remote preflight;
- short-path sync and source verification;
- per-run live logging;
- exact child exit propagation;
- artifact sync-back into repository-owned paths.

  That launcher stabilization is the main durable gain from this campaign.

Canonical Benchmark Impact

This campaign does **not** promote any new canonical paper-cell closure into `Track 1`.

So the canonical benchmark reading remains unchanged:

- no new Tables `2-5` cells close because of this campaign;
- no new Table `6` harmonic becomes fully matched;
- the latest result is informative, but not promotable as a closure upgrade.

Main Conclusions

The campaign supports four conclusions.

1. The Remote Exact-Paper Campaign Path Is Now Operational

The LAN launcher is no longer the blocker. The campaign completed `4/4`, streamed logs correctly, and synchronized all repository-owned artifacts back to the local clone.

2. The Paper-Faithful SVR Grid Does Not Close The Residual Targets

Residual harmonics `40`, `240`, and `162` all remain above the paper target after the recovered `GridSearchCV` path is applied.

3. Isolated SVR Retuning Does Not Beat The Existing Known Values

The isolated reruns do not uncover a better `SVR` configuration than the already observed exact-paper values. This strongly suggests that the remaining gap is structural rather than a launch-orchestration artifact.

4. The Remaining Open Item Has Shifted To ONNX Export

Training and validation now run correctly on the LAN node, but `SVR` ONNX export still fails in all four scoped runs. The deployment-facing export path remains the next exact-paper blocker if `SVR` export parity is still a goal.

Produced Campaign Artifacts

The campaign output folder now contains the required serialized winner artifacts:

- `output/training_campaigns/track1_svr_reference_grid_search_repair_campaign_2026_04_14_22_53_48/campaign_leaderboard.yaml`
- `output/training_campaigns/track1_svr_reference_grid_search_repair_campaign_2026_04_14_22_53_48/campaign_best_run.yaml`
- `output/training_campaigns/track1_svr_reference_grid_search_repair_campaign_2026_04_14_22_53_48/campaign_best_run.md`

The remote wrapper log for the successful campaign execution is:

- `.temp/remote_training_campaigns/2026-04-17-00-54-15_track1_svr_reference_grid_search_repair_campaign_2026_04_14_22_53_48/remote_training_campaign.log`

The canonical reports refreshed by this close-out are:

- `doc/reports/analysis/RCIM Paper Reference Benchmark.md`
- `doc/reports/analysis/Training Results Master Summary.md`